

Name: Kumar, Yulia

ORCID iD: <https://orcid.org/0000-0002-7621-2734>

Position Title: Senior Lecturer

Primary Organization and Location: Kean University, Union, NJ, USA

Synergistic Activities:

- 1. Integration of Research into Education.** Developed and taught undergraduate and graduate computer science courses and independent studies that integrate current research in large language models, transformers, multimodal learning, agentic AI, retrieval-augmented generation, computer vision, data visualization, and responsible AI into lectures, laboratories, programming assignments, and student research projects.
- 2. Mentoring and Broadening Participation in Computing.** Mentors undergraduate, graduate, and high-school researchers, including students from groups underrepresented in computing, through independent studies, capstone projects, and research collaborations, while providing sustained academic advisement to transfer students. Student mentees have coauthored peer-reviewed publications, presented at national conferences and research competitions, and earned first- and second-place poster awards at Great Minds in STEM.
- 3. Advancement of Research Tools and Interdisciplinary Methods.** Develops and disseminates auditable, reproducible methods and student-accessible systems in adversarial LLM testing, bias and cyberbullying detection, data poisoning, retrieval-augmented generation, multimodal AI, quantum natural language processing, data compression, computer vision, and accessible STEM education through collaborations involving Kean, Rutgers, Wenzhou-Kean, and external research partners.
- 4. Scholarly Service and Research Dissemination.** Serves on the AAAI-26 Program Committee, reviews for IEEE, Springer Nature, and other journals and conferences, and serves as a guest editor for special issues on real-time video analytics and AI-driven motion prediction. As a Senior Member of IEEE and a member of ACM and ACM-W, contributes to peer review, research quality, and professional knowledge exchange.
- 5. Institutional and Community STEM Leadership.** Completed three years of service on Kean University's University Curriculum and Distance Learning Committee and supports curriculum development, student advisement, and interdisciplinary AI education. Serves as co-chair of the NCWIT North New Jersey chapter, judges regional STEM fairs, participates in university outreach, and supports the establishment of IEEE student activities at Kean University.